

the security. This is because small blocks and/or small keys can make the ciphertext vulnerable to cryptanalysis attacks.

Of course, you should also know that as the block size and key length increases in length, the cipher process slows down. All CPUs are designed to work with a certain amount of data per cycle. For example, 32-bit computers work with 32 bits per cycle, while 64-bit CPUs can process 64 bits in one cycle (max). So when you make the block size larger than that, each block would need multiple cycles to be transformed and that will slow down the entire process. We'll be using a 32-bit key and a 32-bit block size.

Step 2

Since we're creating a process which can scramble and unscramble messages using the same key, we obviously need to find a mathematical process that's reversible. Step 2 is choosing that method to serve as one of the key parts of the algorithm. For this example, we're choosing a simple 'XOR' method. The table will illustrate the XOR method:

TABLE 1: HOW XOR WORKS

First bit	1	0	1	0
Second bit	0	0	1	1
XOR Result	1	0	0	1

NOTE:

We're going to obtain the bits on the 'second bit' row from the 32-bit key we've decided but we won't use that directly. The reason we would do that is because it makes the whole algorithm just an XOR operation which is too easy to reverse (using a trial and error method) if the key is known to the other person (we're designing a symmetric algorithm, remember?). We're going to make it a little more complicated.

The idea is simple: You take two bits. If both bits are the same (either both 1s or both 0s) then the result is 0. If they're different, the result is a 1. Notice that if you reverse the order of the rows and read bottom up (treat the last row as the First Bit and the First Bit as the XOR Result row) even then the results hold true. This is the simplest way to know that the process is easily reversible. In our example, the first bit row is the actual data, and the second bit would be the key. As long as you know the key (Second Bit), you can encrypt and decrypt.

Step 3

As we said earlier, small block-and key sizes are a bad combination that